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ELECTIVE COURSE-2.4

URA741: INDUSTRY 4.0 IMPLEMENTATION IN PRODUCTION SYSTEM				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: Learn More about the core technologies and development of the 4th Industrial Revolution and how these changes are impacting our personal and professional lives. Learn the basics of efficiency and optimization strategies for a technical system, setting up OPC UA communication in the industrial environment, the Manufacturing Execution System - MES for short. In numerous practical tasks, you will work on the basics of production planning and control using the example of MES4 on the MPS400 system.				
Syllabus Industry 4.0: Introduction to Industry 4.0, Automation pyramid, Energy efficiency in production, fundamentals of energy efficiency, energy flows and reference data, energy efficiency measures in production, commissioning of energy efficiency systems, Basics of Network Technology, IT Security for production systems, implementation and monitoring of security measures in production plant, commissioning of networking devices, protection of PLC, switching and monitoring, security risks of unencrypted access protocols, monitor different security process using PC, Industrial communication with OPCUA, OPCUA client-server communication, Industrial communication with profinet, profibus and profinet basics, profinet in production: setup a network, maintenance, adapt network, trouble shooting of network, ring topology. Plant control and commissioning with MES: fundamentals of production planning and production control, basics of production planning and control, plant control and commissioning with MES, MES4 communication interface to the controller, Smart production control, MES in production, production data and control with MES4, processing and transport times, plant monitoring, value stream analysis and mapping, Webshop customer view for Tracking order status in MES, webshop admin view, optimization of MES process,				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

Plant Visualization with HMI: basics of human machine interface, setup and function, creating HMI device configuration, configure basic objects and elements in TIA portal, development of an HMI for controlling a station module, HMI programming on the stacking magazine and conveyor models.

Object identification and product memory in RFID: Fundamentals of object identification, comparison of identification technologies, selecting an RFID system, RFID in production Automation, control of an RFID system with a PLC, risks and limitations of RFID technology.

Laboratory Work

1. Planning and product control of the production system using MES
2. Plant visualization with HMI
3. Implementation of monitoring and security measures for a production process
4. Autonomous Workpiece feeding on the production process using Mobile Robot
5. Understanding the fundamentals of energy efficiency, energy flows and reference data to optimize the energy consumption of the production system.
6. OPC-UA client server communication
7. Programming Smart sensors to identify workpieces.
8. Perform E-commerce operation using Web shop application.
9. Optimization of production process
10. Performing the planning and process control of MPS 400 system using Ciro simulation software.

Course Learning Outcomes (CLOs)

The students will be able to:

1. interpret traffic captures by using different types of filters and analytics functions, know about security risks of unencrypted access protocols, can monitor and eavesdrop unencrypted messages on the own PC.
2. create a new MPS system in MES4, can create a new product, can create routings and can interpret errors in the process sequence, are able to manually determine the processing and transport times on the MPS system.

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